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MOTION-ACTIVATED SOUND PLAYER

Abstract

A motion-activated sound player is provided. In embodiments, the motion-activated sound player includes a motion sensor assembly and a temperature sensor. When the motion sensor assembly detects motion and the temperature sensor determines an operating temperature is within a predetermined operating temperature range, the motion-activated sound player plays one or more audio tracks. In further embodiments, the motion-activated sound player includes temperature elements, such as heating elements and heat sinks, to maintain the operating temperature within the predetermined operating temperature range. In an example, the motion-activated sound player is installed at an outdoor memorial structure, such as a stone monument (e.g., a gravestone).

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Background/Summary

BACKGROUND

[0001] Audio players are used to project sound in many environments. While often used indoors, audio players may also be installed for use in outdoor, uncovered environments. However, use of audio players in outdoor environments has limitations. For example, audio players in outdoor environments may experience increased wear due to uncontrolled weather conditions. Batteries or other electrical components in the audio player may operate outside of an acceptable operating temperature range due to overly high or overly low outdoor temperatures, damaging the batteries or other electrical components. Audio players face further challenges in outdoor environments when installed in an uncovered setting. In uncovered, outdoor environments, audio players face moisture in the form of rain and snow that can leak into the audio players and damage electrical components.

SUMMARY

[0002] In general terms, this disclosure is directed to a motion-activated sound player. In some embodiments, and by non-limiting example, the motion-activated sound player plays audio when a visitor is detected proximate to the motion-activated sound player and an operating temperature is determined to be within a predetermined operating temperature range.

[0003] In a first aspect, a motion-activated sound player embeddable within a cavity of an outdoor memorial structure is provided. The motion-activated sound player includes a power source, a speaker, a motion sensor assembly, and a controller. The power source supplies power to the motion-activated sound player. The controller includes a temperature sensor and a memory device that stores one or more audio tracks. In response to the controller determining that a visitor is proximate to the motion-activated sound player based on signals from the motion sensor assembly and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.

[0004] In a second aspect, a motion-activated sound player is provided. The motion-activated sound player comprises a capsule, a motion sensor assembly, and a solar panel assembly. The capsule includes a cavity divider, a face plate, and an end cap. The cavity divider defines a first cavity and a second cavity in the capsule. The first cavity houses a speaker, and the second cavity houses a controller and a battery. The controller includes a memory device that stores one or more audio tracks and a temperature sensor. The battery supplies power to the motion-activated sound player. The face plate is attachable to a front face of the capsule to enclose the first cavity, and the end cap is attachable to the capsule to enclose the second cavity. The solar panel assembly converts sunlight to electricity to selectively recharge the battery. In response to the motion sensor assembly detecting motion and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.

[0005] In a third aspect, an audio-enabled stone monument is provided. The audio-enabled stone monument comprises a stone monument and a motion-activated sound player. The motion-activated sound player includes a capsule installed in a bore on a side face of the stone monument, a motion sensor assembly installed on a front face of the stone monument, and a solar panel assembly installed on a top face of the stone monument. The capsule includes a cavity divider defining a first cavity and a second cavity in the capsule. The first cavity houses a speaker, and the second cavity houses a controller and a battery. The controller includes a memory device that stores one or more audio tracks and a temperature sensor. The battery supplies power to the motion-activated sound player. The solar panel assembly converts sunlight to electricity to selectively recharge the battery. In response to the controller determining that a visitor is proximate to the stone monument based on signals from the motion sensor assembly and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.

[0006] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** illustrates an example environment in which aspects of the present disclosure may be implemented.

[0008] FIG. **2** illustrates a schematic block diagram of an example embodiment of a motion-activated sound player.

[0009] FIG. **3** illustrates a schematic block diagram of an example embodiment of a controller for a motion-activated sound player.

[0010] FIG. **4** illustrates an example embodiment of a motion-activated sound player arranged within a capsule installed at a stone monument.

[0011] FIG. **5** illustrates a second example embodiment of a motion-activated sound player arranged within a capsule installed at a stone monument.

[0012] FIG. **6** illustrates a third example embodiment of a motion-activated sound player arranged within a capsule installed at a stone monument.

[0013] FIG. **7** illustrates a fourth example embodiment of a motion-activated sound player arranged within a capsule installed at a stone monument.

[0014] FIG. **8** illustrates an isometric view of a capsule for a motion-activated sound player.

[0015] FIG. **9** illustrates an isometric view of a capsule for a motion-activated sound player.

[0016] FIG. **10** illustrates an isometric view of a capsule for a motion-activated sound player with a face plate.

[0017] FIG. **11** illustrates an example of a motion-activated sound player installed at a stone monument.

[0018] FIG. **12** illustrates an example state diagram of a motion-activated sound player.

[0019] FIG. **13** illustrates a flowchart of an example method for playing audio with a motion-activated sound player.

[0020] FIG. **14** illustrates a flowchart of an example method for modifying settings of a motion-activated sound player.

[0021] FIG. **15** illustrates a schematic diagram of an example embodiment of a computing device for modifying settings of the motion-activated sound player.

DETAILED DESCRIPTION

[0022] Various embodiments will be described in detail with reference to the drawings, wherein like reference numerals represent like parts and assemblies throughout the several views. Reference to various embodiments does not limit the scope of the claims attached hereto. Additionally, any examples set forth in this specification are not intended to be limiting and merely set forth some of the many possible embodiments for the appended claims.

[0023] As used herein, the term “including” as used herein should be read to mean “including, without limitation,” “including but not limited to,” or the like. The term “substantially” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art (and is not to be limited to a special or customized meaning), and furthermore refers without limitation to being largely but not necessarily wholly that which is specified.

[0024] As briefly described above, embodiments, of the present disclosure are directed to a motion-activated sound player. In example aspects, the motion-activated sound player includes a motion sensor assembly, a speaker, and a controller. When the motion sensor assembly detects motion, the controller sends signals to the speaker to play audio. In further aspects, the controller includes a temperature sensor. The temperature sensor monitors an operating temperature of the motion-activated sound player or specific components of the sound player. If the temperature sensor determines that the operating temperature is outside a predetermined operating temperature range, the controller prevents audio playback. By monitoring the operating temperature of the motion-activated sound player and its components to prevent operation outside of the predetermined operating temperature range, the sound player may experience less wear and operate for a longer lifespan.

[0025] In further aspects, the motion-activated sound player includes temperature elements to maintain the operating temperature within the predetermined operating range. For example, a heating coil selectively produces heat to increase the operating temperature to maintain the operating temperature above a lower limit of the predetermined operating range. In another example, a heat sink dissipates heat from electrical components of the motion-activated sound player to cool the electrical components and maintain the operating temperature below an upper limit of the predetermined operating range.

[0026] In further example aspects, the motion-activated sound player includes a capsule enclosing electrical components of the motion-activated sound player. The capsule protects the electrical components from moisture in an outdoor environment. In some examples, the capsule defines multiple cavities. By defining multiple cavities, some electrical components have additional layers of protection from external moisture.

[0027] In further example aspects, the motion-activated sound player is installed at an outdoor memorial structure. In examples, an outdoor memorial structure could include a stone monument, such as a gravestone. In such examples, the motion-activated sound player plays audio tracks when visitors visit the stone monument.

[0028] While examples disclosed herein describe the motion-activated sound player installed at a stone monument, aspects of this disclosure are not limited to such installations. The motion-activated sound player described herein can be installed in other environments while functioning in substantially the same manner. For example, the motion-activated sound player described herein may be installed in other outdoor environments—such as in a mailbox—or in indoor environments—such as in a doorway.

[0029] Turning now to FIG. **1**, an example environment **10** in which aspects of the present disclosure may be implemented is shown. In the illustrated example, a motion-activated sound player **100** is installed in a stone monument **20**. In the illustrated embodiment, the stone monument **20** is a gravestone. In alternative embodiments, the stone monument **20** is any type of outdoor memorial structure. When the motion-activated sound player **100** detects the presence of a visitor **30** proximate to the stone monument **20**, the motion-activated sound player **100** activates to play an audio track.

[0030] FIG. **2** illustrates a schematic block diagram of an example embodiment of the motion-activated sound player **100**. In the illustrated embodiment, the motion-activated sound player **100** includes a controller **200**, a battery **106**, a speaker **108**, temperature elements **110**, a solar panel assembly **112**, and a motion sensor assembly **114**. The controller **200**, the battery **106**, the speaker **108**, and the temperature elements **110** are enclosed within a capsule **300**, and the solar panel assembly **112** and the motion sensor assembly **114** are external to the capsule **300**. In embodiments, such as is shown in FIG. **1**, the motion-activated sound player **100** is installed at a stone monument **20**.

[0031] The controller **200** controls the motion-activated sound player **100**. The controller **200** determines when the motion-activated sound player **100** plays audio tracks. In examples, the controller **200** uses the motion sensor assembly **114** to determine when a visitor is proximate to the motion-activated sound player **100**. When a visitor is detected proximate to the motion-activated sound player **100**, the controller **200** activates the speaker **108** to play an audio track. As described herein, the controller **200** includes a memory storing one or more audio tracks, and the controller **200** selects an audio track to play through the speaker **108** from the memory.

[0032] In further examples, the controller **200** includes a temperature sensor **204**, and the controller **200** controls the motion-activated sound player **100** based on an operating temperature sensed by the temperature sensor **204**. In embodiments, the temperature sensor **204** determines the operating temperature by measuring a temperature within the capsule **300**. In alternative embodiments, the temperature sensor **204** determines the operating temperature by monitoring a temperature of one or more components within the motion-activated sound player **100**. For example, the temperature sensor **204** monitors a temperature of the battery **106**. In other examples, the temperature sensor additionally or alternatively monitors the temperature of other components of the motion-activated sound player **100**, such as the speaker **108** or the controller **200**. The controller **200** compares the operating temperature sensed by the temperature sensor **204** to a predetermined operating temperature range, and if the operating temperature is outside of the predetermined operating temperature range, the controller **200** prevents the motion-activated sound player **100** from playing audio tracks, even if the motion sensor assembly **114** detects a visitor proximate to the motion-activated sound player **100**. In alternative embodiments, the controller **200** additionally or alternatively prevents the battery **106** from charging or discharging when the operating temperature is outside of the predetermined operating temperature range. In an example, the predetermined operating temperature range is −20° C. to +50° C. As described herein, the predetermined operating temperature range is customizable in some embodiments.

[0033] The temperature elements **110** help maintain the operating temperature within the predetermined operating range. Examples of temperature elements **110** are described further herein, particularly with reference to FIGS. **5-7**.

[0034] The speaker **108** plays audio for the motion-activated sound player **100**. In embodiments, the speaker **108** is waterproof, substantially protecting the speaker **108** from water damage when the motion-activated sound player **100** is installed in an uncovered, outdoor environment. In an embodiment, the speaker **108** is an AQ-SPK 3.0UN-4 speaker manufactured by Aquatic AV with its principal place of business in San Jose, CA.

[0035] The battery **106** and the solar panel assembly **112** act as a power source for the motion-activated sound player **100**. The battery **106** supplies power to the controller **200** and, in turn, the other electrical components of the motion-activated sound player **100** such as the speaker **108** and the motion sensor assembly **114**. In an example embodiment, the battery **106** is a rechargeable Lithium-ion battery. The capacity of the battery **106** can be changed depending on the power consumption requirements of the other electrical components in the motion-activated sound player **100** and a desired minimum operating time of the motion-activated sound player **100**. In an example, the motion-activated sound player **100** is configured to have a minimum operating time of 4 hours and the speaker **108** has an average energy consumption of 2 watt-hours, so to accommodate this energy consumption the battery **106** has a capacity greater than 8 watt-hours. The operating voltage of the battery **106** can also be changed based on voltage requirements of the other electrical components of the motion-activated sound player **100**. In an example, the battery **106** has an operating voltage between approximately 3 volts and approximately 4.2 volts, with a nominal voltage of approximately 3.7 volts.

[0036] The solar panel assembly **112** recharges the battery **106**. The solar panel assembly **112** converts sunlight into electricity to recharge the battery **106**. In an embodiment, the controller **200** controls the charging of the battery **106**, and the electricity output by the solar panel assembly **112** returns to the battery **106** through the controller **200**. In alternative embodiments, the solar panel assembly **112** is directly connected with the battery **106**. An output voltage of the solar panel assembly **112** can vary based on a number of cells in the solar panel assembly **112**. In an example, each cell in the solar panel assembly outputs between approximately 0.4 volts and 0.6 volts. The number of cells in the solar panel assembly **112**, an in turn the output voltage of the solar panel assembly **112**, can be configured based on a voltage tolerance of the controller **200** and other electronic components of the motion-activated sound player **100**. In an embodiment, the solar panel assembly **112** has an output voltage between approximately 6 volts and approximately 24 volts.

[0037] As previously described, the motion sensor assembly **114** senses for visitors proximate to the motion-activated sound player **100**. In example embodiments, the motion sensor assembly **114** is a passive infrared sensor. In an example, the motion sensor assembly **114** is a EKMC1601112 passive infrared sensor manufactured by Panasonic Holdings Corporation with its principal place of business in Kadoma, Osaka, Japan. In an embodiment, the motion sensor assembly **114** has a rated detection range of approximately 4 meters and a rated detection angle range of approximately 90° (±45° from vertical). The detection range of the motion sensor assembly **114** can be modified for different operating conditions. For example, when installed at a gravestone, the detection range may be modified to be smaller to avoid detection of individuals that are not visiting the gravestone at which the motion-activated sound player **100** is installed. In an embodiment, the motion sensor assembly **114** is programmed to detect motion out to 6 feet with an angle range of ±25°.

[0038] In an embodiment, the motion sensor assembly **114** is a passive infrared sensor that includes two slots, and each slot is sensitive to infrared radiation. In such an embodiment, the output from the motion sensor assembly **114** is a differential between the two slots. For example, when the motion sensor assembly **114** is idle—i.e., there is no movement-both slots detect the same amount of infrared radiation, and the output signal is approximately zero. When a body producing infrared radiation, such as a human, enters the range monitored by the motion sensor assembly **114**, a first slot of the two slots initially receives more infrared radiation, causing a positive differential to be output by the motion sensor assembly **114**. When the body leaves the range monitored by the motion sensor assembly **114**, a second slot of the two slots receives more infrared radiation, causing a negative differential to be output by the motion sensor assembly **114**. Conversely, a negative differential can be output when a body enters the range of the motion sensor assembly **114**, and a positive differential can be output when the body leaves the range of the motion sensor assembly **114**. As described further herein, in embodiments, the controller **200** uses the positive and negative differentials output from the motion sensor assembly **114** to determine if motion detected by the motion sensor assembly **114** is a visitor of the stone monument **20** at which the motion-activated sound player **100** is installed or a passerby.

[0039] FIG. **3** illustrates a block diagram of an example embodiment of the controller **200**. In the illustrated embodiment, the controller **200** includes a microcontroller **202**, a temperature sensor **204**, an audio driver circuit **206**, a memory **208**, a volume potentiometer **210**, a charge control circuit **212**, a voltage regulator **214**, a motion sensor port **218**, a solar panel port **220**, a battery port **222**, a speaker port **224**, and a programming port **226**. In an embodiment, the controller **200** is a printed circuit board assembly.

[0040] The microcontroller **202** controls the operation of the controller **200** and the various components within the controller **200**. In embodiments, the microcontroller **202** includes a processor configured to execute data instructions stored in the memory **208**.

[0041] As previously described, the temperature sensor **204** determines an operating temperature of a motion-activated sound player. In an embodiment, the operating temperature is a temperature of a component of the motion-activated sound player, such as a battery, a speaker, or the controller **200**.

[0042] The controller **200** operates differently based on the operating temperature sensed by the temperature sensor **204**. In an embodiment, the microcontroller **202** compares the operating temperature to a predetermined operating temperature range to determine operation of the controller **200**. In an example, the microcontroller **202** prevents operation of a speaker if the operating temperature is outside of the predetermined operating temperature range. In another embodiment, the temperature sensor **204** also controls operation of the charge control circuit **212**.

In an example, if the sensed operating temperature is outside of the predetermined operating temperature range, the charge control circuit **212** prevents charging of a battery. In alternative embodiments, the temperature sensor **204** does not control the charge control circuit **212** directly, and instead the microcontroller **202** controls the charge control circuit **212** based on the operating temperature sensed by the temperature sensor **204**.

[0043] The audio driver circuit **206** works in conjunction with the microcontroller **202** to output a signal for a speaker. In an example, the audio driver circuit **206** converts signals from the microcontroller **202** into a format that can be input into the speaker. In another example, the audio driver circuit **206** amplifies audio signals from the microcontroller **202**. In an embodiment, the audio driver circuit **206** includes a class-D amplifier.

[0044] Signals from the audio driver circuit **206** are passed to the speaker port **224** where the signals are then transmitted to the speaker. In an embodiment, the speaker port **224** includes two pins, both of which are used to transmit speaker drive signals. In an example embodiment, the speaker port includes a 3545 Connector manufactured by Keystone Electronics Corp. with its principal place of business in New Hyde Park, NY, and a 3546 Connector also manufactured by Keystone Electronics Corp.

[0045] The memory **208** stores data for the controller **200**. In an example, the memory stores data instructions defining settings for the controller **200**. Examples of settings include the predefined operating temperature range, a maximum playback time, motion sensor ranges, a list of audio tracks playable by the motion-activated sound player, a minimum charge level, a passerby timeframe, a post-playback wait period, and a timeout period length, as described further herein. In another example, the memory **208** stores one or more audio tracks for playback by the motion-activated sound player. In embodiments, the one or more audio tracks are stored in an ordered playlist. In embodiments, the memory **208** also stores data instructions that enables to the motion-activated sound player to perform the functions described herein. In an embodiment, the memory **208** comprises secure digital (SD) flash memory.

[0046] The volume potentiometer **210** controls a playback volume of the motion-activated sound player. In embodiments, the volume potentiometer **210** provides a voltage control input to the microcontroller **202** to define the playback volume of the motion-activated sound player. In an embodiment, the volume potentiometer **210** is a variable resistor. In alternative embodiments, the controller **200** does not include the volume potentiometer **210**, and the playback volume is instead controlled by a programmable playback volume level stored in the memory **208**.

[0047] The charge control circuit **212**, the voltage regulator **214**, the solar panel port **220**, and the battery port **222** make up a power subsystem of the controller **200**. The charge control circuit **212** controls charging of a battery. The charge control circuit **212** receives energy from a solar panel assembly through the solar panel port **220**, and when the charge control circuit **212** determines that the battery needs to be charged, the charge control circuit **212** transfers the energy from the solar panel to the battery port **222**, where the energy is transferred to the battery. In an embodiment, the solar panel port **220** includes two pins: a first pin for a solar panel power supply signal and a second pin for a solar panel power return signal. In an embodiment, the battery port **222** includes two pins: a first pin for a battery positive power terminal and a second pin for a battery negative power terminal. In embodiments, the operation of the charge control circuit **212** is controlled by the temperature sensor **204** or the microcontroller **202** to prevent the charge control circuit **212** from recharging the battery when an operating temperature is outside of a predetermined operating temperature range.

[0048] The voltage regulator **214** supplies a local supply voltage **216** for the controller **200**. The local supply voltage is a steady supply voltage used by the components of the controller **200**. For ease of illustration, the voltage regulator **214** is shown feeding into the local supply voltage **216** rather than illustrating connections between the voltage regulator **214** and each of the other components of the controller **200**. In an embodiment, the voltage regulator **214** draws power from the battery through the battery port **222** to supply the local supply voltage **216**. In alternative embodiments, the voltage regulator **214** additionally or alternatively draws power from the solar panel assembly through the solar panel port **220** and the charge control circuit **212**.

[0049] The motion sensor port **218** connects a motion sensor assembly to the controller **200**. As previously discussed, signals from the motion sensor assembly are sent to the controller **200** when the motion sensor assembly determines that a visitor is proximate to the motion-activated sound speaker, and the controller **200** causes the speaker to play audio. In the illustrated embodiment, signals from the motion sensor port **218** are received by the microcontroller **202**. In an embodiment, the motion sensor port **218** includes connections for three signals between the controller **200** and the motion sensor assembly: a first signal for a motion sensor power supply, a second signal for a motion sensor detection signal, and a third signal for a ground reference. In an example, the second signal for the motion sensor detection signal is a logic signal, and the controller **200** interprets a change in the second signal as an indication that motion was detected.

[0050] As described above, in some embodiments, the motion sensor assembly outputs positive differentials and negative differentials when the motion sensor assembly detects motion from an object entering and exiting the range of the motion sensor assembly. In examples of these embodiments, the controller **200** uses the positive and negative differentials to determine if a visitor is visiting a stone monument at which the motion-activated sound player is installed or if the detected motion is from a passerby. For example, if the controller **200** receives a positive differential signal followed quickly by a negative differential signal, the controller **200** determines the motion sensed by the motion sensor assembly is from a passerby, and the controller **200** does not initiate playback of an audio track. The controller **200** operates similarly if a negative differential signal is followed quickly by a positive differential signal. However, if the controller **200** receives a positive differential signal with no corresponding negative differential signal within a predetermined passerby timeframe—or a negative differential signal with no corresponding positive differential signal within the predetermined passerby timeframe—the controller **200** determines that the motion sensed by the motion sensor assembly is from a visitor, and the controller initiates playback of an audio track. In an example, the passerby timeframe is 3 seconds. In embodiments, the passerby timeframe is customizable.

[0051] In an embodiment, the motion sensor port **218** and the solar panel port **220** are combined in an external connection port. In an example of such an embodiment, the external connection port includes a 5-pin connector: a first pin for a solar panel power supply signal, a second pin for a solar panel power return signal, a third pin for a motion sensor power supply, a fourth pin for a motion sensor detection signal, and a fifth pin for a ground reference. An example of the 5-pin connector includes a IPL1-105-02-L-S-K connector manufactured by Samtec Inc. with its principal place of business in New Albany, IN.

[0052] The programming port **226** allows external computing devices to communicate with the controller **200** to change settings of the motion-activated sound player. In an example, a computing device connected to the controller **200** through the programming port **226** can change a predefined operating temperature range, a maximum playback time, motion sensor ranges, a list of audio tracks playable by the motion-activated sound player, a minimum charge level, a passerby timeframe, a post-playback wait period, and a timeout period length. In an embodiment, the programming port **226** is connected to the microcontroller **202**, and the microcontroller **202** changes settings stored in the memory **208** based on signals from the programming port **226**. In alternative embodiments, the programming port **226** is additionally or alternatively connected to the memory **208**, and signals from the programming port **226** go directly to the memory **208** to modify settings stored in the memory **208**.

[0053] In an embodiment, the programming port **226** is configured to allow a wired connection with an external computing device. In an example embodiment in which the programming port **226** is configured for wired connections, the programming port **226** includes a 10-pin pad field compatible with a TC2050-IDC programming cable manufactured by Tag-Connect LLC with its principal place of business in Burlingame, CA. In this embodiment, the programming port **226** has four data signal pins, a clock signal pin, a ground reference pin, a command signal pin, a power pin, a programming enable pin, and a tenth pin not connected.

[0054] In alternative embodiments, in addition or alternative to the programming port **226**, the controller **200** includes a wireless interface configured to allow wireless communication with a computing device to change settings of the motion-activated sound player. In an example, the wireless interface includes a Bluetooth interface. Like with the programming port **226**, the wireless interface may be connected to the microcontroller **202** and/or the memory **208**.

[0055] FIG. **4** illustrates an example embodiment of the motion-activated sound player **100** arranged within a capsule **300** installed at a stone monument **20**. In the illustrated embodiment, the capsule **300** houses the controller **200**, the battery **106**, and the speaker **108**. The solar panel assembly **112** and the motion sensor assembly **114** are external to the capsule **300** and are connected to the controller **200** via cables extending through the stone monument **20**. The capsule is sealed using an external face plate **308** that is exposed on a face of the stone monument **20** and an end cap **310**. In embodiments, the face plate **308** and the end cap **310** are removable, allowing access to the components within the capsule **300** when removed.

[0056] In the illustrated embodiment, the capsule **300** includes a cavity divider **306** which defines two cavities **302**, **304** within the capsule **300**. In these embodiments, a first cavity **302** houses the speaker **108**, and a second cavity **304** houses the battery **106** and the controller **200**. An aperture **318** (shown in FIGS. **8-9**) in the cavity divider **306** allows the speaker **108** to have a wired connection with the controller **200** despite the speaker **108** and the controller **200** being in different cavities **302**, **304**. The cavity divider **306** provides greater protection for the battery **106** and the controller **200** from moisture by adding an additional layer between the battery **106** and the controller **200** and outdoor conditions that can include rain and snow. In alternative embodiments, the capsule **300** may include multiple cavity dividers **306** defining additional cavities **302**, **304**. Additionally, in alternative embodiments, the components of the motion-activated sound player **100** are arranged in different configurations within the cavities **302**, **304**—e.g., the battery **106** may be in the first cavity **302** with the speaker **108**.

[0057] The capsule **300** also includes standoffs **312** within the second cavity **304**. The standoffs **312** help maintain separation between the battery **106** and the controller **200**. By providing separation between the battery **106** and the controller **200**, the standoffs **312** act as temperature elements, substantially preventing contact between the battery **106** and the controller **200** through which heat could transfer from the battery **106** to the controller **200**, or vice versa, that causes the battery **106** or the controller **200** to overheat.

[0058] FIGS. **5-7** illustrate alterative embodiments of the motion-activated sound player **100** arranged within a capsule **300** installed at a stone monument **20**. The alternative embodiments offer different configurations of the capsule **300** that provide additional temperature elements or enable previously described components to act as temperature elements. Further embodiments may combine elements of the embodiments shown in FIGS. **4-7**.

[0059] FIG. **5** illustrates an example embodiment of the capsule **300** that includes heating elements **314**. In the illustrated embodiment, the heating elements **314** are positioned proximate to the speaker **108**, the battery **106**, and the controller **200** within the capsule **300**. In embodiments, the heating elements **314** are connected to walls of the capsule **300** and the standoffs **312**. In an example embodiment, the heating elements **314** are heating coils, which convert electricity supplied by the controller **200** into heat. In alternative embodiments, the heating elements **314** can be any component capable of converting electricity into heat.

[0060] The heating elements **314** can selectively produce heat to increase the operating temperature of the motion-activated sound player **100**, particularly when the operating temperature is approaching or drops below a lower limit of a predefined operating temperature range. Although not shown for ease of illustration, the heating elements **314** can be electrically connected to and controlled by the controller **200**. When the temperature sensor **204** in the controller **200** determines that the operating temperature is below the lower limit of the predefined operating temperature range—or is approaching the lower limit of the predefined operating temperature range (e.g., is within 5° C. of the lower limit)—the controller **200** activates the heating elements **314** to produce heat, which may cause the operating temperature to increase.

[0061] FIG. **6** illustrates another example embodiment configured to maintain an operating temperature above a lower limit of a predefined operating temperature range. In contrast to the embodiment depicted in FIG. **5**, the embodiment depicted in FIG. **6** does not include additional heating elements **314** and instead uses heat produced by the components of the motion-activated sound player **100**—such as the battery **106**—to increase the operating temperature when necessary. When the temperature sensor **204** determines that the operating temperature is below or approaching the lower limit of the predefined operating temperature range, the controller **200** causes components of the motion-activated sound player **100** to activate, consuming electricity and in turn producing heat. In an example, the controller **200** causes the battery **106** to discharge to produce heat. In another example, the controller **200** uses the speaker **108** to produce heat. In such an example, the controller **200** may cause the speaker **108** to play sounds at a low volume or at a frequency outside of the human audible spectrum such that individuals in the vicinity of the motion-activated sound player **100** do not hear the motion-activated sound player **100** when the speaker **108** is activated for the purposes of producing heat to increase the operating temperature.

[0062] In the illustrated embodiment, the capsule **300** does not include a cavity divider **306** like in the embodiment shown in FIGS. **4** and **5**. Without the cavity divider **306**, heat produced by the components of the motion-activated sound player **100** can more easily increase the operating temperature—e.g., heat produced by the battery **106** can more easily heat up the capsule **300**. In alternative embodiments, the capsule **300** includes the cavity divider **306**.

[0063] FIG. **7** illustrates an example embodiment of the capsule **300** that includes heat sink **316**. The heat sink **316** draws heat away from electrical components of the motion-activated sound player **100** to help maintain the operating temperature below an upper limit of the predetermined operating temperature range. In embodiments, the heat sink **316** transfers heat from the electrical components into the stone monument **20** in which the motion-activated sound player **100** is installed, cooling the electrical components. In an example, the heat sink **316** is made of aluminum. In another example, the heat sink **316** is made of copper. In further examples, the heat sink **316** is made of any metal or other type of material with high thermal conductivity.

[0064] In the illustrated embodiment, the heat sink **316** makes contact with the battery **106** to dissipate heat from the battery **106**. The heat sink **316** extends through the capsule **300** to make contact with the stone monument **20** to transfer the heat from the battery **106** to the stone monument **20**, cooling the battery **106**. In further embodiments, the heat sink **316** additionally or alternatively makes contact with other electrical components in the capsule **300**, such as the speaker **108** and the controller **200**. In an example, the standoffs **312** are part of the heat sink **316** and made of the same material as the heat sink **316**, making contact with the controller **200** to diffuse heat from the controller **200** into the stone monument **20**. In another example, the cavity divider **306** is removed and the heat sink **316** divides the cavities **302**, **304**, allowing the heat sink **316** to make contact with the speaker **108** in the first cavity **302** and the battery **106** and controller **200** in the second cavity **304**.

[0065] Turning to FIGS. **8-10**, an example embodiment of the capsule **300** is shown. In the illustrated embodiment, the capsule **300** is generally cylindrical in shape. In embodiments, the capsule **300** is made substantially of plastic. In alternative embodiments, the capsule **300** is made of any weather and water-resistant material.

[0066] As previously described, the capsule **300** includes a cavity divider **306** to define a first cavity **302** (shown in FIG. **9**) and a second cavity **304**. Standoffs **312** are attached to the cavity divider **306** within the second cavity **304**, which maintain separation between a battery and a controller, as previously described. The cavity divider **306** also includes an aperture **318** through which a wired connection can be maintained between electrical components in the first cavity **302**—such as a speaker—and electrical components in the second cavity **304**—such as a controller. In embodiments, the aperture **318** is a minimum size through which a wired connection is still capable of being maintained. As previously described, the cavity divider **306** helps to protect electrical components in the second cavity **304** from external moisture, such as rain and snow, so having a large aperture **318** may reduce the protection provided by the cavity divider **306**.

[0067] An end cap **310** seals the second cavity **304**. In an embodiment, the end cap **310** is removable, allowing access to the electrical components within the second cavity **304**. In an example embodiment, the end cap **310** includes threads that interact with corresponding threads on the capsule **300** with which the end cap **310** can screw onto the capsule **300**. In another example embodiment, the end cap **310** has a circumference slightly greater than a circumference of the capsule **300**, allowing the end cap **310** to maintain a tight fit with the capsule **300** when pressed onto the capsule **300**.

[0068] In the illustrated embodiment, the end cap **310** includes apertures **320** through which a wired connection can be maintained between electrical components in the second cavity **304**—such as a controller—and external electrical components—such as a solar panel assembly and a motion sensor assembly. Like with the aperture **318**, in embodiments, the apertures **320** are a minimum size through which a wired connection is still capable of being maintained to minimize the amount of moisture that can enter the capsule **300** through the apertures **320**.

[0069] A face plate **308** seals the first cavity **302**. The face plate **308** is made of weather resistant materials, as it is the most exposed component of the motion-activated sound player to weather conditions. In an example embodiment, the face plate **308** is made substantially of stainless steel.

[0070] In embodiments, the face plate **308** includes apertures **328**, allowing sound from a speaker in the first cavity **302** to more easily be heard through the face plate **308**. In the illustrated embodiment, the face plate **308** attaches to a front face **322** of the capsule **300** using fasteners **326**. The fasteners **326** extend through apertures **324** in the front face **322** to attach the face plate **308** to the front face **322**. In an example, the apertures **324** are threaded and the fasteners **326** are screws. In alternative embodiments, other types of mechanical fasteners are used, such as nuts and bolts, nails, or rivets. In further embodiments, other attachment mechanisms are used, including adhesives.

[0071] Referring now to FIG. **11**, an example embodiment of the motion-activated sound player **100** installed in a stone monument **20** is shown. In the illustrated embodiment, the stone monument **20** is a gravestone with a side face **22**, a top face **24**, and a front face **26**. The capsule **300** of the motion-activated sound player **100** is installed into a bore in the side face **22** of the stone monument **20**. The face plate **308** substantially covers the bore, helping to keep moisture out of the bore and away from the capsule **300**. In embodiments, the capsule **300** is removable from the stone monument **20**. The capsule **300** can be removed to allow access for a computing device to establish a wired connection with the motion-activated sound player **100**—for example, a wired connection can be established through a programming port to modify settings of the motion-activated sound player **100**.

[0072] The solar panel assembly **112** is installed on the top face **24** of the stone monument **20**. By installing the solar panel assembly **112** on the top face **24**, the solar panel assembly **112** is exposed to more sunlight and can provide more electricity to the motion-activated sound player **100**. In an embodiment, the solar panel assembly **112** is attached to anchors mounted into the stone monument **20** to attach the solar panel assembly **112** to the top face **24**. In alternative embodiments, the solar panel assembly **112** is attached to the top face **24** using other attachment methods, such as using other fasteners or adhesives.

[0073] The motion sensor assembly **114** is installed on the front face **26** of the stone monument **20**.

By installing the motion sensor assembly **114** on the front face **26**, the motion sensor assembly **114** can detect when visitors approach the front face **26** of the stone monument **20**. In an embodiment, the motion sensor assembly **114** is installed in a bore in the front face **26**, reducing the visibility of the motion sensor assembly **114** from visitors visiting the stone monument **20** and better protecting the motion sensor assembly **114** from weather conditions. In alternative embodiments, the motion sensor assembly **114** is attached to the front face **26** using other attachment methods, such as using fasteners or adhesives.

[0074] The solar panel assembly **112** and the motion sensor assembly **114** have a wired connection with electrical components in the capsule **300**. In the illustrated embodiment, the wired connections between the capsule **300** and the solar panel assembly **112** and the motion sensor assembly **114** are maintained through channels carved through the stone monument **20**. In alternative embodiments, the wired connections between the capsule **300** and the solar panel assembly **112** and the motion sensor assembly **114** are external to the stone monument **20**. In further embodiments, the connection between the motion sensor assembly **114** and the capsule **300** is wireless.

[0075] FIG. **12** illustrates an example state diagram **400** of a motion-activated sound player. In the illustrated embodiment, the motion-activated sound player operates in four states: standby mode **402**, playback mode **404**, timeout mode **406**, and programming mode **408**.

[0076] Standby mode **402** is the default state of the motion-activated sound player. While operating in standby mode **402**, the motion-activated sound player operates in a low power state in which audio is not enabled and the motion-activated sound player is monitoring for nearby motion and connections with external computing devices.

[0077] The motion-activated sound player also monitors an operating temperature while in standby mode **402**. In embodiments, if the operating temperature is determined to be outside of a predetermined operating temperature range or approaching an upper limit or a lower limit of the predetermined operating temperature range (e.g., the operating temperature is within 5° C. of the upper/lower limit), the motion-activated sound player uses temperature elements to maintain the operating temperature within the predetermined operating temperature range. In an example, the motion-activated sound player uses heating coils to increase the operating temperature if the operating temperature is below or approaching the lower limit of the predetermined operating temperature range. In alternative embodiments, the motion-activated sound player uses passive temperature elements, such as a heat sink, to maintain the operating temperature within the predetermined operating temperature range and does not actively trigger the temperature elements in response to the sensed operating temperature being outside or approaching the upper or lower limits of the predetermined operating temperature range.

[0078] As previously described, a motion sensor assembly is used to monitor for motion within a programmed range. When motion is detected by the motion sensor assembly, the motion-activated sound player transitions into playback mode **404**. In some embodiments, as previously described, signals from the motion sensor assembly—such as positive and negative differentials—are used to determine if the motion sensed is from a visitor or a passerby. In such embodiments, the motion-activated sound player transitions into playback mode **404** if the motion is determined to be from a visitor, but not if the motion is determined to be from a passerby.

[0079] In playback mode **404**, the motion-activated sound player plays audio. In an embodiment, the audio played by the motion-activated sound player includes at least a portion of one or more audio tracks. The motion-activated sound player selects an audio track to play from among one or more audio tracks stored in a memory of the motion-activated sound player. In embodiments, the one or more audio tracks are ordered as a playlist, and the motion-activated sound player progresses through the playlist as audio tracks are selected. In an example embodiment, the motion-activated sound player tracks the last audio track that is played at the end of an instance in which the motion-activated sound player is operating in playback mode **404** so that the next time the motion-activated sound player enters playback mode **404**, the next audio track in the playlist is selected—e.g., if the motion-activated sound player ended an instance of operating in playback mode **404** by playing the fourth audio track in the playlist, the fifth audio track in the playlist is selected the next time the motion-activated sound player enters playback mode **404**.

[0080] In embodiments, the motion-activated sound player plays the selected audio track, and after playback of the audio track is complete, the motion-activated sound player monitors for motion for a predetermined post-playback wait period—e.g., two minutes. If motion is detected during the wait period, the motion-activated sound player selects another audio track from the one or more audio tracks for playback. If no motion is detected during the wait period, the motion-activated sound player returns to standby mode **402**. In an embodiment, the motion-activated sound player remains in playback mode **404** until the motion-activated sound player goes through a complete wait period in which no motion is detected, at which point the motion-activated sound player transitions back to standby mode **402**. In embodiments, the post-playback wait period is a setting stored in a memory of the motion-activated sound player that is customizable.

[0081] In an alternative embodiment, the motion-activated sound player remains in playback mode **404** until the earlier of the motion-activated sound player going through a complete wait period in which no motion is detected or a maximum playback time being reached. In this embodiment, if the motion-activated sound player goes through a complete wait period in which no motion is detected, the motion-activated sound player returns to standby mode **402**. If the maximum playback time is reached, the motion-activated sound player transitions into timeout mode **406**. As previously described, the maximum playback time is a setting stored in the memory of the motion-activated sound player which is customizable in some embodiments. In an example, the maximum playback time is two hours. In a further embodiment, the motion-activated sound player remains in playback mode **404** until the earlier of the motion-activated sound player going through a complete wait period in which no motion is detected or a battery level falling below a minimum charge level. In this embodiment, if the motion-activated sound player goes through a complete wait period in which no motion is detected, the motion-activated sound player returns to standby mode **402**. If the battery level falls below the minimum charge level, the motion-activated sound player transitions into timeout mode **406**. As previously described, the minimum charge level is a setting stored in the memory of the motion-activated sound player which is customizable in some embodiments.

[0082] While in the timeout mode **406**, playback of audio is disabled. In embodiments, the motion-activated sound player remains in timeout mode **406** for a predetermined timeout period—e.g., 30 minutes. As previously described, the predetermined timeout period is a setting stored in the memory of the motion-activated sound player which is customizable in some embodiments. By remaining in timeout mode **406** for the predetermined timeout period, the battery of the motion-activated sound player has an opportunity to recharge. After the predetermined timeout period ends, the motion-activated sound player transitions from timeout mode **406** to standby mode **402**.

[0083] From standby mode **402**, the motion-activated sound player can also transition into programming mode **408**. When a connection is established between the motion-activated sound player and an external computing device while the motion-activated sound player is in standby mode **402**, the motion-activated sound player transitions into programming mode **408**. As previously described, the connection between the motion-activated sound player and the external computing device can be wired or wireless, depending on the embodiment of the motion-activated sound player. In alternative embodiments, the motion-activated sound player can transition from playback mode **404** and timeout mode **406** to programming mode **408** when a connection is established with the external computing device.

[0084] While in programming mode **408**, audio playback is disabled and the motion-activated sound player receives programming instructions from the external computing device to change settings of the motion-activated sound player. In examples, the external computing device changes settings related to a predefined operating temperature range, a maximum playback time, motion sensor ranges, a list of audio tracks playable by the motion-activated sound player, a minimum charge level, a passerby timeframe, a post-playback wait period, and a timeout period length. After the settings are customized and the connection with the external computing device is terminated, the motion-activated sound player returns to standby mode **402**.

[0085] In some embodiments, the motion-activated sound player also considers the operating temperature when transitioning between states. In an example, while in standby mode **402**, in addition to monitoring for motion with the motion sensor assembly, the motion-activated sound player also monitors the operating temperature for use in determining whether to transition into playback mode **404**. In these embodiments, if the operating temperature is outside of the predetermined operating temperature range, the motion-activated sound player does not transition into playback mode **404**, even if the motion sensor assembly detects motion. In another example, while in playback mode **404**, if the operating temperature moves outside the predetermined operating temperature range, the motion-activated sound player returns to standby mode **402**.

[0086] FIG. **13** illustrates a flowchart of an example method **500** for playing audio using a motion-activated sound player. The method **500** includes operations **502**, **504**, **506**, **508**, **510**.

[0087] The operation **502** is performed to sense for the presence of a visitor. In embodiments, the presence of a visitor is sensed by monitoring for motion in a predetermined range around the motion-activated sound player. In an example, the operation **502** is performed by a motion sensor assembly. In embodiments, the motion sensor assembly includes a passive infrared sensor.

[0088] The operation **504** is performed to determine if a visitor is detected. In embodiments, if motion is detected during the operation **502**, it is determined that a visitor is detected. In alternative embodiments, motion detected during the operation **502** is differentiated between motion from a visitor and motion from a passerby. In embodiments, the operation **504** is performed by a controller in the motion-activated sound player. In examples in which motion is differentiated between a visitor and a passerby, the controller determines if a positive differential is quickly followed by a negative differential (or vice versa). If a positive differential is followed by a negative differential within a predetermined passerby timeframe, the motion is determined to be from a passerby. If the positive differential is not followed by a negative differential within the predetermined passerby timeframe, the motion is determined to be from a visitor. If a visitor is detected, the method **500** proceeds to the operation **506**. If a visitor is not detected, the method **500** returns to the operation **502**.

[0089] The operation **506** is performed to determine an operating temperature of the motion-activated sound player. In embodiments, the operating temperature is a temperature of one or more components of the motion-activated sound player. In alternative embodiments, the operating temperature is an ambient temperature within the motion-activated sound player. In an example, the operation **506** is performed by a temperature sensor in the motion-activated sound player. In embodiments, the temperature sensor is included in a controller of the motion-activated sound player. In embodiments, the temperature sensor monitors temperatures of one or more components of the motion-activated sound player, such as a battery, the controller, or a speaker.

[0090] The operation **508** is performed to determine if the operating temperature determined during the operation **506** is within a predetermined operating temperature range. By comparing the operating temperature to the predetermined operating temperature range, the motion-activated sound player can be protected from operating in temperatures that can cause damage to components of the motion-activated sound player. In embodiments, the operation **508** is performed by a controller of the motion-activated sound player. In an example embodiment, the predetermined operating temperature range is a setting stored in a memory of the motion-activated sound player and is customizable based on programming by a user. If the operating temperature is not within the predetermined operating temperature range, the method **500** returns to the operation **502**. If the operating temperature is within the predetermined operating temperature range, the method **500** proceeds to the operation **510**.

[0091] The operation **510** is performed to playback an audio track. In an embodiment, a controller selects an audio track from a memory and causes a speaker to play the audio track. In an example, the audio track is part of an ordered playlist of audio tracks, and the audio track is selected for playback based on which audio track of the playlist was most recently played—i.e., the audio track that is selected during the operation **510** is the next audio track in the playlist. In an embodiment, the operation **510** includes transitioning the motion-activated sound player from a standby mode to a playback mode. After playback of the audio track is completed, the method **500** returns to the operation **502**, and the method **500** can loop. In embodiments, when the method **500** returns to the operation **502**, the motion-activated sound player transitions back to the standby mode.

[0092] In alternative embodiments, the method **500** returns to the operation **502** before the audio track is played completely. In an example, if the operating temperature leaves the predetermined operating temperature range during the operation **510**, playback of the audio track ends, and the method returns to the operation **502**. In another example, if a battery level of the motion-activated sound player falls below a predetermined minimum charge level, playback of the audio track ends, and the method returns to the operation **502**.

[0093] The method **500** can include additional or alternative operations in other embodiments. For example, in some embodiments, after playback of the audio track during the operation **510**, the motion-activated sound player compares the current playback time to a maximum playback time, and if the current playback time is greater than or equal to the maximum playback time, the motion-activated sound player enters a timeout mode for a predetermined amount of time, as described above. In another example, if the operating temperature is determined to be outside of the predetermined operating temperature range, the motion-activated sound player uses temperature elements to control the operating temperature—e.g., heating coils are activated to produce heat and increase the operating temperature.

[0094] The operations **502**, **504**, **506**, **508**, **510** can also be performed in a different order than shown in FIG. **13**. For example, in some embodiments, the operations **506**, **508** to determine the operating temperature and compare it to the predetermined operating temperature range are performed before the operations **502**, **504** to sense for visitor presence and determine if a visitor was detected.

[0095] FIG. **14** illustrates a flowchart of an example method **600** for modifying settings of a motion-activated sound player. The method **600** includes operations **602**, **604**, **606**, **608**, **610**, **612**.

[0096] The operation **602** is performed to establish a connection between the motion-activated sound player and a computing device. In an embodiment, a wired connection is established between the devices. In such embodiments, the wired connection is established with a programming port on a controller of the motion-activated sound player. In alternative embodiments, a wireless connection is established between the devices, such as a Bluetooth connection. In such embodiments, the wireless connection is established through a wireless interface in the controller of the motion-activated sound player.

[0097] The operation **604** is performed to disable audio playback by the motion-activated sound player. In an embodiment, a controller of the motion-activated sound player determines that a computing device is connected with the motion-activated sound player and transitions the motion-activated sound player into a programming mode in which playback is disabled.

[0098] The operation **606** is performed to receive programming instructions. The programming instructions are received from the computing device connected to the motion-activated sound player during the operation **602**. In embodiments, the programming instructions describe modifications of settings of the motion-activated sound player, including a predefined operating temperature range, a maximum playback time, motion sensor ranges, a list of audio tracks playable by the motion-activated sound player, a minimum charge level, a passerby timeframe, a post-playback wait period, and a timeout period length. In an example, the programming instructions are received by a controller through a programming port.

[0099] The operation **608** is performed to modify settings of the motion-activated sound player according to the programming instructions received during the operation **606**. In an embodiment, a controller modifies a memory of the motion-activated sound player. In alternative embodiments, the computing device connects directly with the memory of the motion-activated sound player to modify the memory.

[0100] The operation **610** is performed to terminate the connection between the computing device and the motion-activated sound player. After the connection is terminated, the method **600** proceeds to the operation **612** in which playback is reenabled. In an embodiment, reenabling playback includes to transition the motion-activated sound player from the programming mode to a standby mode in which playback is not directly enabled but from which the motion-activated sound player can transition into a playback mode upon detection of a visitor.

[0101] FIG. **15** illustrates a schematic diagram of an example embodiment of a computing device **700** for modifying settings of the motion-activated sound player. In an example, the computing device **700** is a mobile device, such as a smartphone. In other examples, the computing device **700** is a personal computer, such as a laptop. In some embodiments, the computing device **700** forms a wired connection with the motion-activated sound player. In alternative embodiments, the computing device forms a wireless connection, such as a Bluetooth connection, with the motion-activated sound player. In the illustrated embodiment, the computing device **700** includes an input device **702**, an output device **704**, a processor **706**, a wired interface **708**, a wireless interface **710**, a power supply **712**, and a memory **714**.

[0102] The input device **702** operates to receive input from external sources. Such sources can include inputs received from a user. The inputs can be received through a touchscreen, a stylus, a keyboard, etc.

[0103] The output device **704** operates to provide output of information from the computing device **700**. For example, a display can output visual information while a speaker can output audio information.

[0104] The processor **706** reads data and instructions. The data and instructions can be stored locally, received from an external source, or accessed from removable media.

[0105] The wired interface **708** allows for a wired connection with external devices, such as the motion-activated sound player. In an example, the wired interface **708** includes a USB-C port. In alternative embodiments, the wired interface **708** can include any type of port configured for a wired connection, including other USB ports.

[0106] The wireless interface **710** allows for wireless connections with external devices, such as the motion-activated sound player. In an embodiment, the wireless interface **710** includes a Bluetooth Low Energy (BLE) interface. In alternative embodiments, the wireless interface **710** can include an interface for other forms of wireless communication, including Wi-Fi, Zigbee, or Z-wave.

[0107] The power supply **712** provides power to the processor **706**.

[0108] The memory **714** includes software applications **720** and an operating system **722**. The memory **714** contains data and instructions that are usable by the processor to implement various functions of the computing device **700**.

[0109] The software applications **720** can include applications usable to perform various functions on the computing device **700**. One such application is a motion-activated sound player application **724**. In a particular embodiment, when the motion-activated sound player application **724** is operating on the computing device **700**, the motion-activated sound player application **724** can be configured to provide a user interface for modifying settings of a motion-activated sound player to which the computing device **700** is connected, either through the wired interface **708** or the wireless interface **710**.

[0110] The various embodiments described above are provided by way of illustration only and should not be construed to limit the claims attached hereto. Those skilled in the art will readily recognize various modifications and changes that may be made without following the example embodiments and applications illustrated and described herein, and without departing from the full scope of the following claims.

Claims

1. A motion-activated sound player embeddable within a cavity of an outdoor memorial structure, comprising: a power source, the power source supplying power to the motion activated-sound player; a speaker; a motion sensor assembly; and a controller, the controller including: a temperature sensor; and a memory device, the memory device storing one or more audio tracks; wherein in response to the controller determining that a visitor is proximate to the motion-activated sound player based on signals from the motion sensor assembly and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.
2. The motion-activated sound player of claim 1, wherein the power source includes: a battery; and a solar panel assembly.
3. The motion-activated sound player of claim 1, wherein the predetermined operating temperature range is −20° C. to +50° C.
4. The motion-activated sound player of claim 1, wherein the one or more audio tracks are ordered in a playlist.
5. The motion-activated sound player of claim 4, wherein the selected audio track is selected based on a previously played audio track and an ordering of the playlist.
6. The motion-activated sound player of claim 1, wherein the controller further includes a programming port, wherein the programing port received signals from an external computing device to modify one or more settings of the motion-activated sound player.
7. The motion-activated sound player of claim 6, wherein the one or more settings of the motion-activated sound player include one or more of the predefined operating temperature range, a maximum playback time, motion sensor ranges, a list of audio tracks playable by the motion-activated sound player, a minimum charge level, a passerby timeframe, a post-playback wait period, and a timeout period length.
8. The motion-activated sound player of claim 1, further comprising: one or more temperature elements, wherein the one or more temperature elements maintain the operating temperature within the predetermined operating temperature range.
9. The motion-activated sound player of claim 8, wherein the temperature elements include one or more heating coils.
10. The motion-activated sound player of claim 9, wherein in response to the temperature sensor determining that the operating temperature is below the predetermined operating temperature range, the controller activates the heating coils to produce heat.
11. The motion-activated sound player of claim 1, further comprising: a capsule, including: a cavity divider defining a first cavity and a second cavity in the capsule; wherein the speaker is housed in the first cavity and the controller is housed in the second cavity.
12. The motion-activated sound player of claim 11, further comprising: a stone monument; wherein the capsule is installed in a bore in a side face of the stone monument; wherein the motion sensor assembly is installed on a front face of the stone monument; and wherein the power source is at least partially installed on a top face of the stone monument.
13. The motion-activated sound player of claim 12, further comprising: a heat sink, the heat sink contacting the stone monument and at least part of the power source to diffuse heat from the at least part of the power source to the stone monument.
14. The motion-activated sound player of claim 1, wherein the controller determines that the visitor is proximate to the motion-activated sound player based on signals from the motion sensor

assembly if the controller receives, from the motion sensor assembly, a first signal indicating that the visitor entered a motion sensor range without receiving a second signal from the motion sensor assembly indicating that the visitor exited the motion sensor range within a predetermined passerby timeframe.

15. The motion-activated sound player of claim 1, wherein if the controller determines that a playback time has exceeded a maximum playback time, the controller transitions the motion-activated sound player to a timeout mode in which playback of audio tracks is disabled.

16. A motion-activated sound player, comprising: a capsule, including: a cavity divider defining a first cavity and a second cavity in the capsule; a face plate attachable to a front face of the capsule to enclose the first cavity; and an end cap attachable to the capsule to enclose the second cavity; wherein the first cavity houses: a speaker; and wherein the second cavity houses: a controller, including: a memory device, the memory device storing one or more audio tracks; a temperature sensor; and a battery, the battery supplying power to the motion activated-sound player; a motion sensor assembly; and a solar panel assembly, the solar panel assembly converting sunlight to electricity to selectively recharge the battery; wherein in response to the motion sensor assembly detecting motion and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.

17. The motion-activated sound player of claim 16, wherein if the temperature sensor determines that the operating temperature is not within the predetermined operating temperature range, the controller prevents recharging of the battery by the solar panel assembly.

18. The motion-activated sound player of claim 17, further comprising: a stone monument; wherein the capsule is installed in a bore in a side face of the stone monument; wherein the motion sensor assembly is installed on a front face of the stone monument; and wherein the solar panel assembly is installed on a top face of the stone monument.

19. An audio-enabled stone monument, comprising: a stone monument; and a motion-activated sound player, including: a capsule installed in a bore on a side face of the stone monument, the capsule including: a cavity divider defining a first cavity and a second cavity in the capsule; wherein the first cavity houses: a speaker; and wherein the second cavity houses: a controller, including: a memory device, the memory device storing one or more audio tracks; a temperature sensor; and a battery, the battery supplying power to the motion activated-sound player; a motion sensor assembly installed on a front face of the stone monument; and a solar panel assembly installed on a top face of the stone monument, the solar panel assembly converting sunlight to electricity to selectively recharge the battery; wherein in response to the controller determining that a visitor is proximate to the stone monument based on signals from the motion sensor assembly and the temperature sensor determining that an operating temperature is within a predetermined operating temperature range, the controller sends a signal to the speaker to play at least a portion of a selected audio track of the stored one or more audio tracks.

20. The audio-enabled stone monument of claim 19, wherein the stone monument is a gravestone.